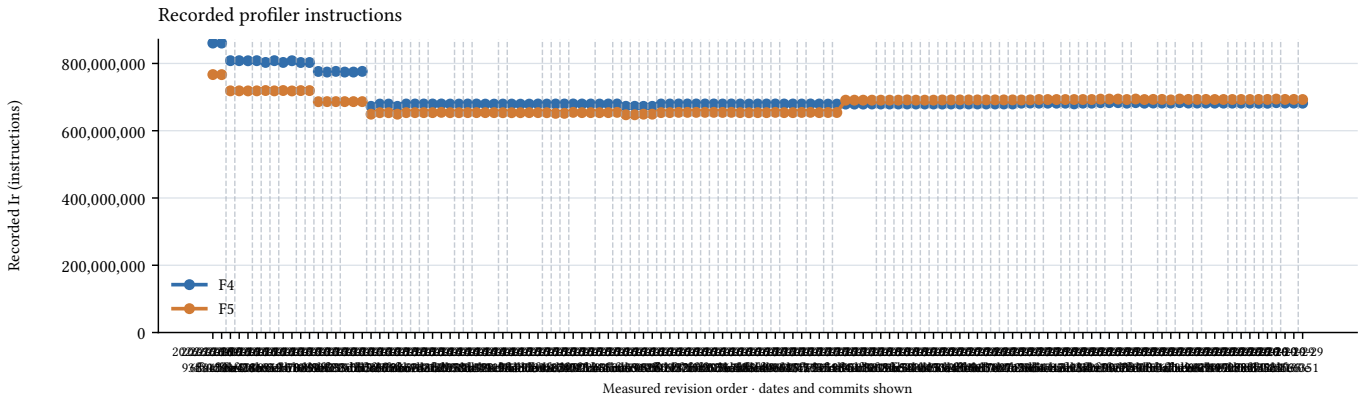
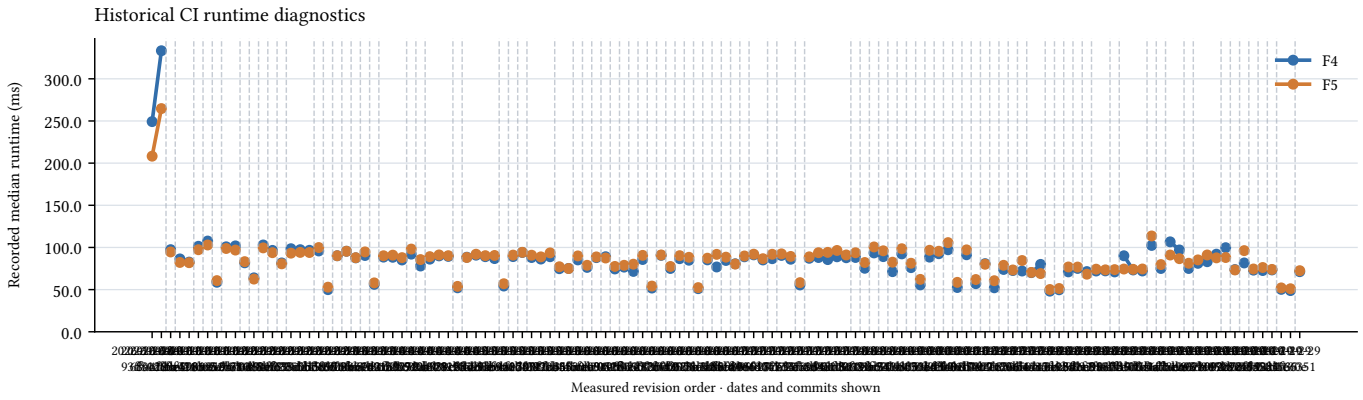
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




125 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.




125 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |
| 2026-10-12 / 89e3233c | 776,490,367 | 686,543,371 | 81.628 | 80.581 |
| 2026-10-12 / 80f5391d | 774,379,185 | 686,519,896 | 98.501 | 93.239 |
| 2026-10-12 / c62555b7 | 774,379,185 | 686,519,896 | 97.360 | 94.117 |
| 2026-10-12 / 633d2158 | 776,467,358 | 686,512,000 | 96.637 | 94.008 |
| 2026-10-12 / ca0d5f86 | 672,500,776 | 649,174,334 | 95.848 | 99.895 |
| 2026-10-12 / b5cc6ffc | 679,242,389 | 653,622,849 | 49.965 | 52.882 |
| 2026-10-13 / bf92906b | 679,242,389 | 653,622,849 | 90.163 | 90.201 |
| 2026-10-13 / d85756aa | 672,486,486 | 649,160,075 | 95.497 | 95.480 |
| 2026-10-13 / ab28b57b | 679,449,939 | 653,823,121 | 87.697 | 87.878 |
| 2026-10-13 / 56f66e43 | 679,449,939 | 653,828,331 | 90.448 | 94.715 |
| 2026-10-13 / 28b1daf2 | 679,449,939 | 653,828,331 | 56.215 | 57.961 |
| 2026-10-14 / cc962159 | 679,449,939 | 653,831,656 | 88.546 | 90.149 |
| 2026-10-14 / 33816db5 | 679,549,997 | 654,627,509 | 88.344 | 90.874 |
| 2026-10-14 / 191a0739 | 679,240,609 | 653,664,759 | 85.135 | 87.917 |
| 2026-10-14 / cd226d0c | 679,240,609 | 653,664,759 | 92.059 | 98.073 |
| 2026-10-14 / b2272a44 | 679,449,939 | 653,873,401 | 78.007 | 85.621 |
| 2026-10-14 / bc82d128 | 679,449,939 | 653,873,401 | 86.126 | 89.102 |
| 2026-10-14 / 912a98ed | 679,098,623 | 653,830,968 | 89.995 | 91.125 |
| 2026-10-14 / 8d8b1295 | 679,098,623 | 653,830,968 | 89.395 | 90.230 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / b339e847 | 679,240,608 | 653,621,583 | 52.150 | 53.839 |
| 2026-10-15 / 7912faa5 | 679,226,318 | 653,606,027 | 88.406 | 88.091 |
| 2026-10-15 / 3c5d435e | 678,905,208 | 653,631,022 | 90.855 | 91.956 |
| 2026-10-15 / f0599bf9 | 679,098,622 | 653,825,706 | 89.060 | 90.253 |
| 2026-10-15 / 3db0ecbc | 679,449,938 | 653,830,913 | 86.813 | 90.492 |
| 2026-10-15 / addb6a02 | 679,256,525 | 653,632,330 | 54.248 | 56.951 |
| 2026-10-15 / 12348e50 | 679,449,939 | 651,714,930 | 89.150 | 90.941 |
| 2026-10-15 / b906df02 | 679,449,939 | 651,714,179 | 94.066 | 94.227 |
| 2026-10-15 / cf274521 | 679,549,984 | 654,627,503 | 88.061 | 90.795 |
| 2026-10-15 / 1b874fcc | 679,449,939 | 653,830,975 | 86.246 | 88.831 |
| 2026-10-15 / 700f7d31 | 679,444,579 | 653,824,679 | 89.023 | 93.545 |
| 2026-10-15 / 289bb5fb | 679,264,149 | 653,687,604 | 74.802 | 77.102 |
| 2026-10-16 / 92246530 | 679,449,938 | 653,829,966 | 75.128 | 75.009 |
| 2026-10-16 / 3a4a5cec | 679,549,995 | 654,621,347 | 85.229 | 89.944 |
| 2026-10-16 / 333c08c0 | 672,605,656 | 647,427,799 | 76.341 | 78.831 |
| 2026-10-16 / 4cfcb94e | 672,605,656 | 647,427,799 | 87.944 | 88.781 |
| 2026-10-16 / ca0310ff | 672,486,486 | 649,202,105 | 89.012 | 87.501 |
| 2026-10-16 / de0c9391 | 672,500,776 | 649,216,408 | 74.498 | 77.487 |
| 2026-10-16 / 7051f4e2 | 679,449,938 | 653,873,407 | 76.675 | 78.834 |
| 2026-10-16 / 082d5507 | 679,445,034 | 653,868,507 | 71.580 | 80.148 |
| 2026-10-16 / 2f02e32c | 679,549,984 | 654,627,503 | 85.711 | 90.622 |
| 2026-10-16 / 6d437f0c | 679,549,984 | 654,627,503 | 51.852 | 54.190 |
| 2026-10-16 / 6bf2bcf0 | 679,549,997 | 654,621,861 | 90.619 | 91.074 |
| 2026-10-16 / f8300065 | 679,549,997 | 654,669,949 | 75.064 | 77.431 |
| 2026-10-16 / ccf43056 | 679,549,997 | 654,669,949 | 86.686 | 90.254 |
| 2026-10-16 / 57b028d8 | 679,350,531 | 654,421,882 | 84.863 | 88.297 |
| 2026-10-16 / f25d5e52 | 679,549,984 | 654,669,936 | 51.138 | 52.369 |
| 2026-10-16 / 213e7a80 | 679,449,939 | 653,830,971 | 85.272 | 87.253 |
| 2026-10-16 / 78adfda4 | 679,240,609 | 653,621,067 | 76.846 | 91.802 |
| 2026-10-17 / b1bf5b89 | 679,449,939 | 653,825,765 | 84.571 | 88.476 |
| 2026-10-17 / 40bcf690 | 679,449,939 | 653,825,765 | 80.639 | 80.305 |
| 2026-10-17 / 8efd4e48 | 679,549,984 | 654,622,293 | 88.930 | 89.875 |
| 2026-10-17 / 89036527 | 679,449,939 | 653,830,971 | 91.230 | 91.994 |
| 2026-10-17 / a90646f1 | 679,435,649 | 653,817,360 | 85.170 | 86.662 |
| 2026-10-17 / b59187e5 | 679,340,625 | 654,418,146 | 86.682 | 91.890 |
| 2026-10-17 / fd043d4f | 679,549,955 | 654,621,012 | 90.548 | 92.367 |
| 2026-10-17 / 17c71374 | 679,449,939 | 653,830,397 | 86.043 | 89.344 |
| 2026-10-17 / 7d751c8d | 679,098,491 | 654,182,814 | 55.523 | 58.269 |
| 2026-10-17 / a95fcd95 | 679,098,493 | 654,182,133 | 87.157 | 88.860 |
| 2026-10-17 / 2c1164df | 679,198,639 | 690,798,036 | 88.066 | 93.755 |
| 2026-10-17 / e59342d3 | 679,198,639 | 690,798,036 | 85.604 | 94.177 |
| 2026-10-17 / ca00c151 | 679,198,639 | 690,798,036 | 88.918 | 96.426 |
| 2026-10-18 / f8501268 | 679,449,938 | 690,826,836 | 87.801 | 91.097 |
| 2026-10-18 / a1d0f1e6 | 679,449,938 | 690,826,836 | 88.118 | 93.700 |
| 2026-10-18 / c8613fe5 | 679,449,938 | 690,825,106 | 75.312 | 82.144 |
| 2026-10-18 / 3c192c34 | 679,449,938 | 690,825,106 | 93.514 | 100.589 |
| 2026-10-18 / 805b6279 | 679,549,993 | 691,623,334 | 89.125 | 96.021 |
| 2026-10-19 / 21d553b1 | 679,453,166 | 690,824,740 | 71.297 | 82.510 |
| 2026-10-19 / fc5d4835 | 679,449,937 | 690,826,048 | 92.362 | 98.482 |
| 2026-10-20 / 841a45cd | 679,451,718 | 690,805,442 | 76.066 | 81.266 |
| 2026-10-20 / 0e3c51ed | 679,340,663 | 691,396,123 | 55.582 | 62.124 |
| 2026-10-20 / d0337ff5 | 679,551,773 | 691,602,405 | 88.546 | 96.658 |
| 2026-10-20 / d30a63b8 | 679,549,994 | 691,600,618 | 92.571 | 95.491 |
| 2026-10-20 / 892ef408 | 679,549,994 | 691,600,618 | 97.275 | 105.731 |
| 2026-10-21 / b4f05610 | 679,551,773 | 691,602,032 | 52.495 | 58.611 |
| 2026-10-21 / 43e01b10 | 679,551,773 | 691,602,032 | 91.335 | 97.133 |
| 2026-10-22 / 02b37807 | 679,551,773 | 691,602,032 | 57.057 | 61.964 |
| 2026-10-23 / 6ed792e8 | 679,549,983 | 691,604,220 | 80.788 | 80.079 |
| 2026-10-23 / d34b9b39 | 679,549,983 | 691,604,220 | 52.168 | 60.548 |
| 2026-10-27 / 7e271266 | 681,943,949 | 691,347,680 | 73.901 | 78.829 |
| 2026-10-27 / c402a154 | 682,282,779 | 691,707,737 | 72.619 | 73.294 |
| 2026-10-27 / 1c756bcb | 681,899,901 | 692,697,266 | 72.047 | 84.493 |
| 2026-10-28 / a5db1413 | 681,899,901 | 692,697,266 | 70.151 | 70.376 |
| 2026-10-28 / 0650ca8b | 681,899,901 | 692,697,266 | 79.912 | 69.027 |
| 2026-10-28 / d4e85a6a | 681,885,611 | 692,687,520 | 48.174 | 50.139 |
| 2026-10-28 / e74711b4 | 679,810,476 | 692,710,161 | 49.796 | 51.316 |
| 2026-10-28 / cafd2389 | 681,943,647 | 692,740,867 | 71.045 | 76.833 |
| 2026-10-28 / eec5dcc3 | 681,943,647 | 692,740,867 | 75.109 | 76.743 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-28 / 98a2f7fa | 683,245,465 | 693,927,563 | 71.387 | 68.293 |
| 2026-10-28 / 3f0e13ee | 683,415,446 | 694,093,622 | 72.074 | 74.215 |
| 2026-10-28 / 7be0960b | 683,415,446 | 694,093,622 | 72.232 | 73.439 |
| 2026-10-28 / a988ff46 | 681,943,879 | 692,741,225 | 70.987 | 73.705 |
| 2026-10-28 / 8177cd08 | 683,437,564 | 694,188,206 | 90.137 | 74.414 |
| 2026-10-28 / 0d839d5b | 681,899,901 | 692,698,574 | 73.298 | 74.059 |
| 2026-10-28 / 7fe71c4a | 682,335,928 | 693,119,282 | 72.101 | 74.442 |
| 2026-10-29 / 25d0e671 | 682,335,928 | 693,119,282 | 102.392 | 113.613 |
| 2026-10-29 / 0fc7cdad | 682,218,335 | 691,617,426 | 75.044 | 79.697 |
| 2026-10-29 / 3f1dcd9e | 683,439,590 | 694,127,041 | 106.645 | 91.019 |
| 2026-10-29 / 0af11c62 | 682,263,636 | 693,066,268 | 97.025 | 86.703 |
| 2026-10-29 / f73e667b | 682,263,636 | 693,066,268 | 75.045 | 81.275 |
| 2026-10-29 / be2969ce | 682,263,636 | 693,066,268 | 81.162 | 85.238 |
| 2026-10-29 / ca0c809d | 682,037,440 | 692,830,863 | 83.099 | 91.075 |
| 2026-10-29 / 85d80508 | 682,037,440 | 692,830,863 | 92.024 | 87.672 |
| 2026-10-29 / 91902d0c | 682,037,440 | 692,830,863 | 99.730 | 88.072 |
| 2026-10-29 / 117c9c4f | 682,076,130 | 692,882,707 | 73.768 | 73.314 |
| 2026-10-29 / 992c2579 | 682,032,151 | 692,835,720 | 81.606 | 96.247 |
| 2026-10-29 / 2c583c5c | 681,686,676 | 692,985,843 | 73.089 | 74.436 |
| 2026-10-29 / 78f84234 | 681,686,676 | 692,980,550 | 72.621 | 76.212 |
| 2026-10-29 / 8eb04d5e | 683,400,900 | 694,082,886 | 73.332 | 73.948 |
| 2026-10-29 / 055c5035 | 682,218,705 | 693,065,470 | 50.283 | 52.017 |
| 2026-10-29 / 11b1130e | 682,218,705 | 693,065,470 | 48.863 | 51.023 |
| 2026-10-29 / 21f66351 | 681,905,560 | 692,750,765 | 71.271 | 72.453 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

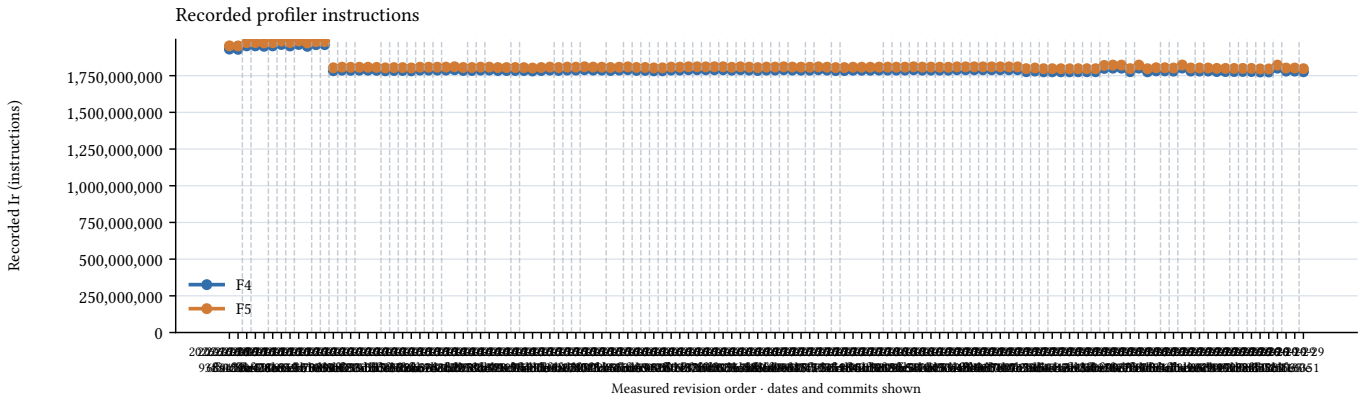
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

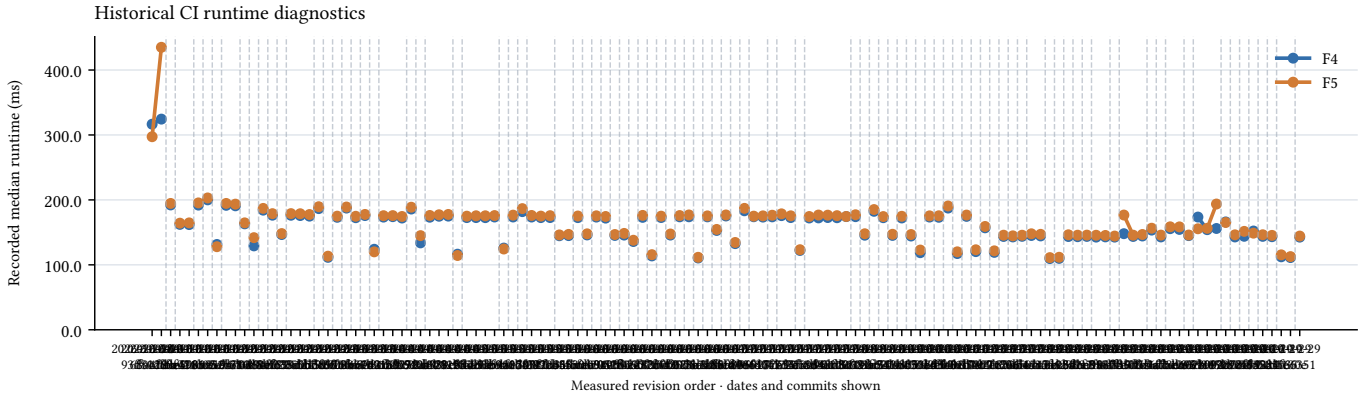
Evidence: 125 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

# Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.




125 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.




125 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 1,930,989,200 | 1,953,108,992 | 316.570 | 297.297 |
| 2026-10-09 / 6f81260f | 1,929,374,570 | 1,952,598,745 | 324.403 | 435.004 |
| 2026-10-11 / 8a41eca7 | 1,953,021,023 | 1,974,789,730 | 192.203 | 194.561 |
| 2026-10-11 / 0cfba411 | 1,953,020,766 | 1,975,267,516 | 162.266 | 164.161 |
| 2026-10-11 / f88e5909 | 1,949,981,977 | 1,973,219,339 | 161.982 | 164.584 |
| 2026-10-11 / fb926c02 | 1,953,020,766 | 1,974,789,781 | 191.988 | 195.374 |
| 2026-10-11 / e478b952 | 1,962,376,386 | 1,985,422,672 | 200.058 | 202.996 |
| 2026-10-11 / aa0cee7b | 1,952,402,641 | 1,975,976,609 | 131.376 | 127.933 |
| 2026-10-12 / 81ca1c78 | 1,962,376,386 | 1,985,933,163 | 191.630 | 194.567 |
| 2026-10-12 / a8194d61 | 1,949,694,281 | 1,972,772,104 | 190.899 | 193.455 |
| 2026-10-12 / 5cbe16d5 | 1,961,124,313 | 1,981,687,200 | 163.135 | 164.619 |
| 2026-10-12 / eb70862e | 1,961,124,313 | 1,981,686,883 | 129.169 | 141.847 |
| 2026-10-12 / 31088bfc | 1,783,643,232 | 1,803,568,510 | 184.204 | 186.912 |
| 2026-10-12 / 0d34f782 | 1,787,314,371 | 1,807,797,314 | 176.437 | 178.824 |
| 2026-10-12 / 89e3233c | 1,786,351,448 | 1,806,842,750 | 146.640 | 148.159 |
| 2026-10-12 / 80f5391d | 1,787,314,371 | 1,807,244,283 | 176.353 | 178.878 |
| 2026-10-12 / c62555b7 | 1,787,314,371 | 1,807,244,283 | 175.899 | 178.689 |
| 2026-10-12 / 633d2158 | 1,786,234,678 | 1,806,168,901 | 174.990 | 177.494 |
| 2026-10-12 / ca0d5f86 | 1,783,289,117 | 1,802,994,551 | 186.580 | 189.492 |
| 2026-10-12 / b5cc6ffc | 1,785,257,434 | 1,804,956,249 | 111.497 | 113.333 |
| 2026-10-13 / bf92906b | 1,785,257,434 | 1,804,956,249 | 173.194 | 174.965 |
| 2026-10-13 / d85756aa | 1,783,003,746 | 1,802,715,389 | 187.409 | 189.039 |
| 2026-10-13 / ab28b57b | 1,787,965,641 | 1,807,670,514 | 172.579 | 174.785 |
| 2026-10-13 / 56f66e43 | 1,787,969,850 | 1,807,674,723 | 175.850 | 177.627 |
| 2026-10-13 / 28b1daf2 | 1,787,969,850 | 1,807,674,723 | 123.983 | 119.998 |
| 2026-10-14 / cc962159 | 1,787,965,641 | 1,808,168,023 | 173.510 | 175.690 |
| 2026-10-14 / 33816db5 | 1,790,194,799 | 1,810,456,555 | 174.166 | 175.821 |
| 2026-10-14 / 191a0739 | 1,785,256,826 | 1,805,459,208 | 172.076 | 174.475 |
| 2026-10-14 / cd226d0c | 1,785,256,826 | 1,805,459,208 | 185.930 | 188.735 |
| 2026-10-14 / b2272a44 | 1,787,965,186 | 1,808,230,025 | 133.809 | 144.964 |
| 2026-10-14 / bc82d128 | 1,787,965,186 | 1,808,230,025 | 173.338 | 176.053 |
| 2026-10-14 / 912a98ed | 1,784,912,427 | 1,805,177,722 | 174.838 | 177.121 |
| 2026-10-14 / 8d8b1295 | 1,784,912,427 | 1,805,177,722 | 175.021 | 177.613 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / b339e847 | 1,785,255,750 | 1,805,524,217 | 116.412 | 114.358 |
| 2026-10-15 / 7912faa5 | 1,784,970,955 | 1,804,687,873 | 172.752 | 174.937 |
| 2026-10-15 / 3c5d435e | 1,782,202,991 | 1,801,914,239 | 172.763 | 175.551 |
| 2026-10-15 / f0599bf9 | 1,784,990,781 | 1,805,255,060 | 172.667 | 175.531 |
| 2026-10-15 / 3db0ecbc | 1,787,964,110 | 1,808,232,577 | 173.668 | 175.897 |
| 2026-10-15 / addb6a02 | 1,785,268,499 | 1,805,533,338 | 125.807 | 124.378 |
| 2026-10-15 / 12348e50 | 1,788,056,225 | 1,808,318,933 | 173.678 | 176.547 |
| 2026-10-15 / b906df02 | 1,788,056,225 | 1,807,750,349 | 181.927 | 186.574 |
| 2026-10-15 / cf274521 | 1,790,285,878 | 1,810,547,645 | 173.613 | 176.055 |
| 2026-10-15 / 1b874fcc | 1,787,976,859 | 1,808,241,709 | 172.732 | 175.034 |
| 2026-10-15 / 700f7d31 | 1,787,976,859 | 1,807,671,995 | 172.808 | 175.713 |
| 2026-10-15 / 289bb5fb | 1,785,371,563 | 1,805,638,717 | 144.794 | 146.086 |
| 2026-10-16 / 92246530 | 1,788,058,429 | 1,807,748,501 | 144.989 | 146.889 |
| 2026-10-16 / 3a4a5cec | 1,790,297,890 | 1,809,989,439 | 172.711 | 175.194 |
| 2026-10-16 / 333c08c0 | 1,785,727,004 | 1,805,891,517 | 146.016 | 147.750 |
| 2026-10-16 / 4cfcb94e | 1,785,727,004 | 1,805,891,517 | 173.541 | 175.495 |
| 2026-10-16 / ca0310ff | 1,783,097,959 | 1,802,805,840 | 171.952 | 174.458 |
| 2026-10-16 / de0c9391 | 1,783,382,754 | 1,803,084,509 | 145.240 | 146.698 |
| 2026-10-16 / 7051f4e2 | 1,788,058,429 | 1,808,325,583 | 145.816 | 148.406 |
| 2026-10-16 / 082d5507 | 1,788,058,429 | 1,808,325,583 | 135.905 | 137.873 |
| 2026-10-16 / 2f02e32c | 1,790,289,796 | 1,810,551,563 | 173.111 | 176.139 |
| 2026-10-16 / 6d437f0c | 1,790,289,796 | 1,810,551,563 | 113.741 | 115.682 |
| 2026-10-16 / 6bf2bcf0 | 1,790,220,548 | 1,809,884,826 | 172.852 | 174.962 |
| 2026-10-16 / f8300065 | 1,790,220,548 | 1,810,482,315 | 145.826 | 147.493 |
| 2026-10-16 / ccf43056 | 1,790,220,548 | 1,810,482,315 | 173.926 | 175.928 |
| 2026-10-16 / 57b028d8 | 1,787,512,707 | 1,807,204,760 | 173.993 | 176.851 |
| 2026-10-16 / f25d5e52 | 1,790,210,949 | 1,810,472,716 | 110.513 | 111.595 |
| 2026-10-16 / 213e7a80 | 1,787,980,594 | 1,808,241,908 | 173.897 | 175.384 |
| 2026-10-16 / 78adfda4 | 1,785,272,234 | 1,804,971,060 | 152.949 | 154.518 |
| 2026-10-17 / b1bf5b89 | 1,788,059,441 | 1,808,326,595 | 175.225 | 176.824 |
| 2026-10-17 / 40bcf690 | 1,788,059,441 | 1,808,326,595 | 132.645 | 134.442 |
| 2026-10-17 / 8efd4e48 | 1,790,210,430 | 1,810,472,197 | 183.238 | 186.980 |
| 2026-10-17 / 89036527 | 1,788,059,441 | 1,808,322,547 | 174.065 | 174.959 |
| 2026-10-17 / a90646f1 | 1,787,695,280 | 1,807,903,799 | 173.534 | 175.412 |
| 2026-10-17 / b59187e5 | 1,787,474,342 | 1,807,737,121 | 173.891 | 176.406 |
| 2026-10-17 / fd043d4f | 1,790,182,702 | 1,809,892,450 | 175.761 | 178.537 |
| 2026-10-17 / 17c71374 | 1,787,980,075 | 1,807,678,901 | 172.788 | 175.501 |
| 2026-10-17 / 7d751c8d | 1,784,810,930 | 1,805,015,067 | 122.130 | 123.306 |
| 2026-10-17 / a95fcd95 | 1,784,812,954 | 1,805,078,260 | 172.140 | 174.437 |
| 2026-10-17 / 2c1164df | 1,787,129,943 | 1,807,393,178 | 172.286 | 176.688 |
| 2026-10-17 / e59342d3 | 1,787,129,943 | 1,807,393,178 | 172.825 | 176.495 |
| 2026-10-17 / ca00c151 | 1,787,129,943 | 1,807,393,178 | 172.532 | 175.739 |
| 2026-10-18 / f8501268 | 1,787,980,091 | 1,808,180,618 | 174.301 | 174.497 |
| 2026-10-18 / a1d0f1e6 | 1,787,980,091 | 1,808,180,618 | 174.167 | 177.133 |
| 2026-10-18 / c8613fe5 | 1,787,980,091 | 1,807,674,211 | 145.609 | 147.871 |
| 2026-10-18 / 3c192c34 | 1,787,980,091 | 1,807,674,211 | 182.414 | 185.143 |
| 2026-10-18 / 805b6279 | 1,790,216,532 | 1,810,414,822 | 172.598 | 174.347 |
| 2026-10-19 / 21d553b1 | 1,787,695,296 | 1,807,902,799 | 145.303 | 147.070 |
| 2026-10-19 / fc5d4835 | 1,787,979,079 | 1,808,242,913 | 172.170 | 174.760 |
| 2026-10-20 / 841a45cd | 1,787,979,207 | 1,807,773,273 | 144.351 | 146.669 |
| 2026-10-20 / 0e3c51ed | 1,787,512,766 | 1,807,304,761 | 118.682 | 122.765 |
| 2026-10-20 / d0337ff5 | 1,790,216,532 | 1,809,990,751 | 173.574 | 175.443 |
| 2026-10-20 / d30a63b8 | 1,790,222,138 | 1,809,996,357 | 173.214 | 175.560 |
| 2026-10-20 / 892ef408 | 1,790,222,138 | 1,809,996,357 | 187.412 | 190.624 |
| 2026-10-21 / b4f05610 | 1,790,217,528 | 1,810,009,523 | 117.362 | 120.020 |
| 2026-10-21 / 43e01b10 | 1,790,221,126 | 1,810,013,121 | 174.595 | 176.646 |
| 2026-10-22 / 02b37807 | 1,790,217,528 | 1,810,009,523 | 120.350 | 123.189 |
| 2026-10-23 / 6ed792e8 | 1,790,210,446 | 1,809,979,585 | 156.998 | 159.094 |
| 2026-10-23 / d34b9b39 | 1,790,210,446 | 1,809,979,585 | 119.174 | 121.524 |
| 2026-10-27 / 7e271266 | 1,776,988,613 | 1,797,707,009 | 143.678 | 145.644 |
| 2026-10-27 / c402a154 | 1,781,495,997 | 1,802,211,486 | 142.967 | 144.644 |
| 2026-10-27 / 1c756bcb | 1,776,237,685 | 1,797,266,192 | 143.898 | 145.607 |
| 2026-10-28 / a5db1413 | 1,776,237,685 | 1,797,266,192 | 145.191 | 147.819 |
| 2026-10-28 / 0650ca8b | 1,776,237,685 | 1,797,266,192 | 144.515 | 146.993 |
| 2026-10-28 / d4e85a6a | 1,775,956,559 | 1,796,991,899 | 109.618 | 111.144 |
| 2026-10-28 / e74711b4 | 1,776,473,583 | 1,797,499,723 | 109.894 | 111.658 |
| 2026-10-28 / cafd2389 | 1,776,720,989 | 1,797,369,864 | 143.591 | 146.283 |
| 2026-10-28 / eec5dcc3 | 1,776,720,989 | 1,797,369,864 | 143.314 | 145.658 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-28 / 98a2f7fa | 1,799,185,931 | 1,819,795,133 | 143.748 | 145.769 |
| 2026-10-28 / 3f0e13ee | 1,801,087,270 | 1,822,006,681 | 142.700 | 145.559 |
| 2026-10-28 / 7be0960b | 1,801,087,270 | 1,822,006,681 | 143.045 | 145.262 |
| 2026-10-28 / a988ff46 | 1,776,956,800 | 1,797,817,362 | 142.697 | 144.302 |
| 2026-10-28 / 8177cd08 | 1,801,324,894 | 1,822,152,664 | 147.968 | 176.686 |
| 2026-10-28 / 0d839d5b | 1,776,189,713 | 1,797,398,940 | 143.690 | 145.684 |
| 2026-10-28 / 7fe71c4a | 1,782,794,781 | 1,803,971,012 | 144.406 | 146.744 |
| 2026-10-29 / 25d0e671 | 1,782,794,781 | 1,803,971,012 | 153.481 | 156.380 |
| 2026-10-29 / 0fc7cdad | 1,780,841,978 | 1,801,871,192 | 142.949 | 145.781 |
| 2026-10-29 / 3f1dcd9e | 1,801,870,248 | 1,822,542,227 | 155.684 | 158.554 |
| 2026-10-29 / 0af11c62 | 1,781,483,875 | 1,802,227,857 | 154.518 | 158.454 |
| 2026-10-29 / f73e667b | 1,781,483,875 | 1,802,227,857 | 144.992 | 145.908 |
| 2026-10-29 / be2969ce | 1,781,483,875 | 1,802,227,857 | 173.576 | 155.521 |
| 2026-10-29 / ca0c809d | 1,778,150,479 | 1,799,321,272 | 154.165 | 156.253 |
| 2026-10-29 / 85d80508 | 1,778,150,479 | 1,799,321,272 | 155.985 | 193.637 |
| 2026-10-29 / 91902d0c | 1,778,150,479 | 1,799,321,272 | 166.071 | 165.173 |
| 2026-10-29 / 117c9c4f | 1,778,781,973 | 1,799,496,223 | 142.918 | 146.213 |
| 2026-10-29 / 992c2579 | 1,778,186,668 | 1,799,215,980 | 143.946 | 151.537 |
| 2026-10-29 / 2c583c5c | 1,775,115,354 | 1,795,822,710 | 152.087 | 148.766 |
| 2026-10-29 / 78f84234 | 1,775,115,354 | 1,795,822,710 | 143.839 | 146.346 |
| 2026-10-29 / 8eb04d5e | 1,801,040,010 | 1,822,022,220 | 143.310 | 145.363 |
| 2026-10-29 / 055c5035 | 1,781,181,074 | 1,801,920,997 | 112.354 | 115.382 |
| 2026-10-29 / 11b1130e | 1,781,181,074 | 1,801,920,997 | 111.183 | 112.679 |
| 2026-10-29 / 21f66351 | 1,776,548,381 | 1,797,443,384 | 142.655 | 144.301 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

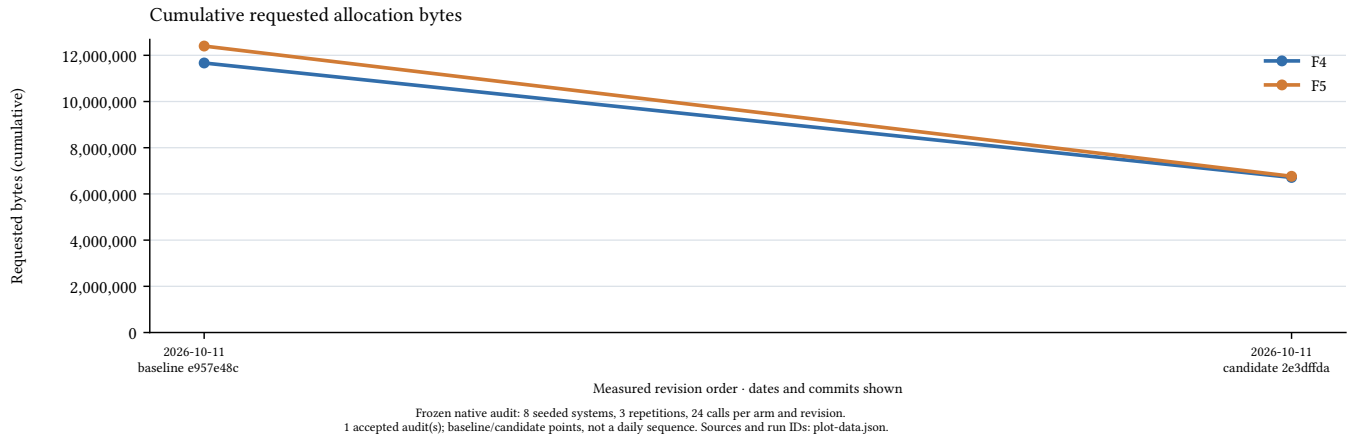
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

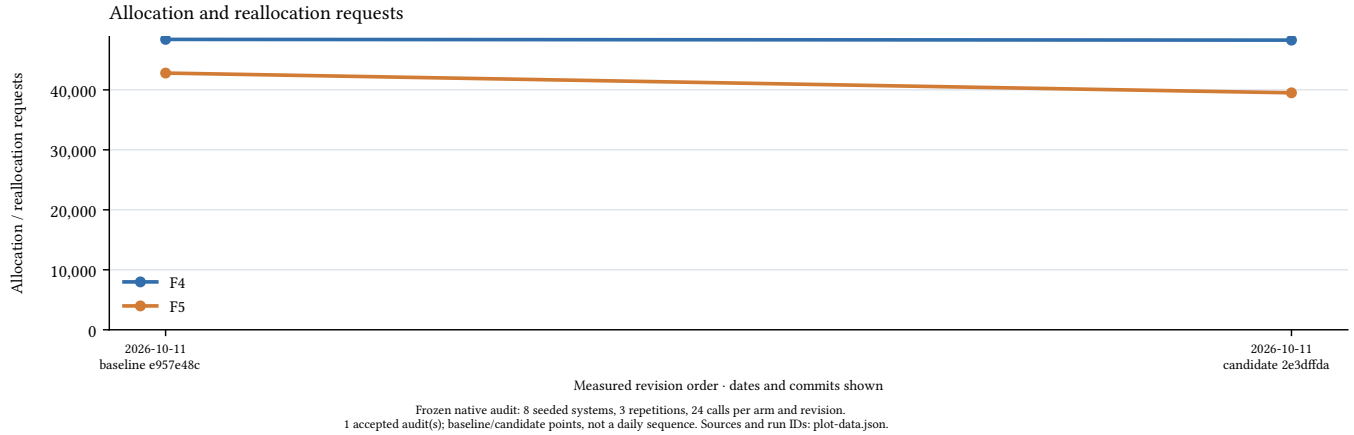
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 125 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

## Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F4 bytes | F5 bytes | F4 requests | F5 requests |
|---|---|---|---|---|
| 2026-10-11 / e957e48c | 11,667,696 | 12,400,920 | 48,414 | 42,801 |
| 2026-10-11 / 2e3dffda | 6,712,608 | 6,762,696 | 48,294 | 39,507 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

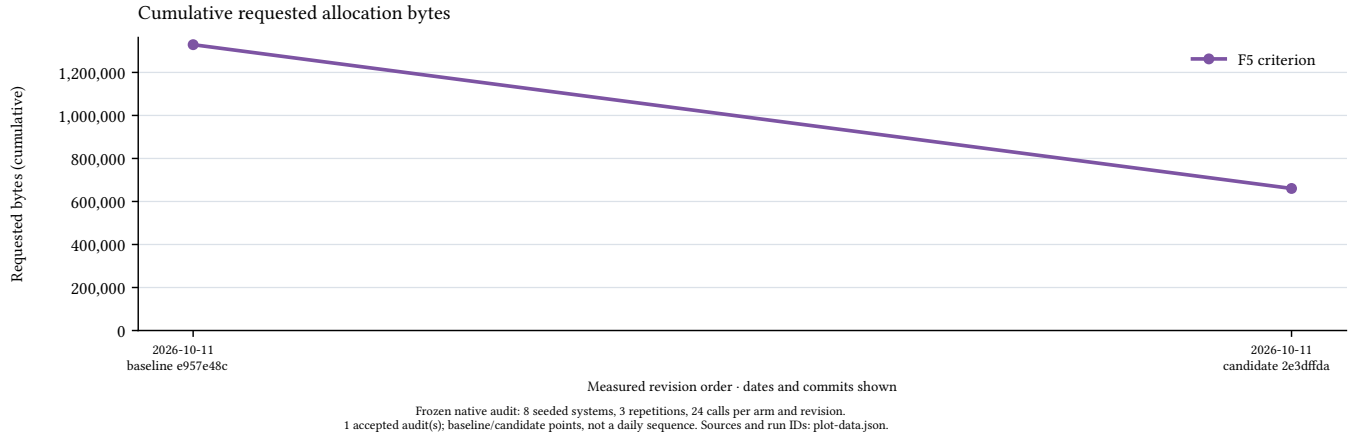
Criterion construction is a separate part of F5 and is never compared as a complete solver.

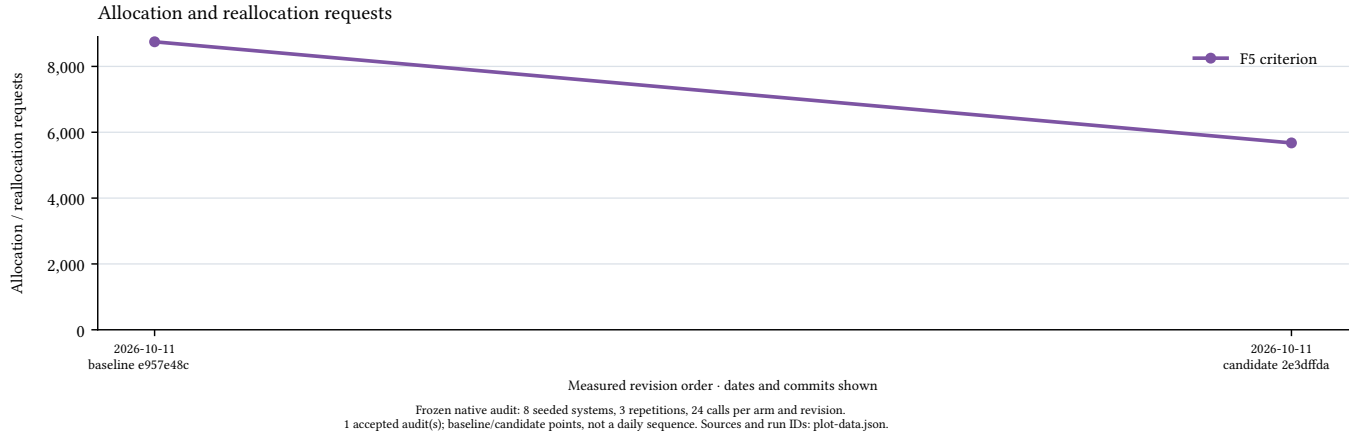
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

## F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F5 criterion bytes | F5 criterion requests |
|---|---|---|
| 2026-10-11 / e957e48c | 1,329,024 | 8,748 |
| 2026-10-11 / 2e3dffda | 660,480 | 5,676 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.